# Project Design Phase-I

**SolutionArchitecture**

| | |
|---|---|
| Date | 8 March 2026 |
| Team ID | Team 230 |
| Project Name | Insurance Fraud Detection Using Machine Learning |
| Maximum Marks | NA |

**Solution Architecture:**

**1. Data Collection Layer**
Insurance claim datais collected from multiple sources such as customer databases, claim forms, vehicle/property records, medical billing systems, and historical insurance datasets.

**2. Data Storage Layer**
Thecollected dataisstored in a centralized database or data warehouse where structured and historical claim data can be securely maintained for further analysis.

**3. Data Preprocessing Layer**
Rawdata iscleaned andprepared by handling missing values, removing duplicates, encoding categorical variables, and normalizing numerical features to make the dataset suitable for machine learning models.

**4. Feature Engineering Layer**
Important features suchasclaim amount, claim frequency, customer history, accident details, billing patterns, and property value are extracted to improve fraud detection accuracy.

**5. Machine Learning Model Layer**
Machine learning algorithms suchas Logistic Regression, Decision Tree, Random Forest, or Gradient Boosting are used to train a fraud detection model that classifies claims as fraudulent or genuine.

**6. Fraud Detection Engine**
Thetrained model analyzes incoming insurance claims in real-time and predicts the probability of fraud based on learned patterns**.**

**7. Application / API Layer**
AwebapplicationorRESTAPI allows insurance companies to submit claims data

and receive fraud prediction results from the system.

**8. Monitoring and Reporting Layer**
Dashboards and reports display suspicious claims, fraud probability scores, and analytics to help investigators and risk managers make data-driven decisions.

**9. Security and Access Control**
Authentication, authorization, and data encryption mechanisms ensure that sensitive insurance data is protected and accessible only to authorized users.

**10.DeploymentLayer**
The system is deployed on cloud platforms or enterprise servers where it can scale to handle large volumes of insurance claims efficiently.

**Example - Solution Architecture Diagram:**

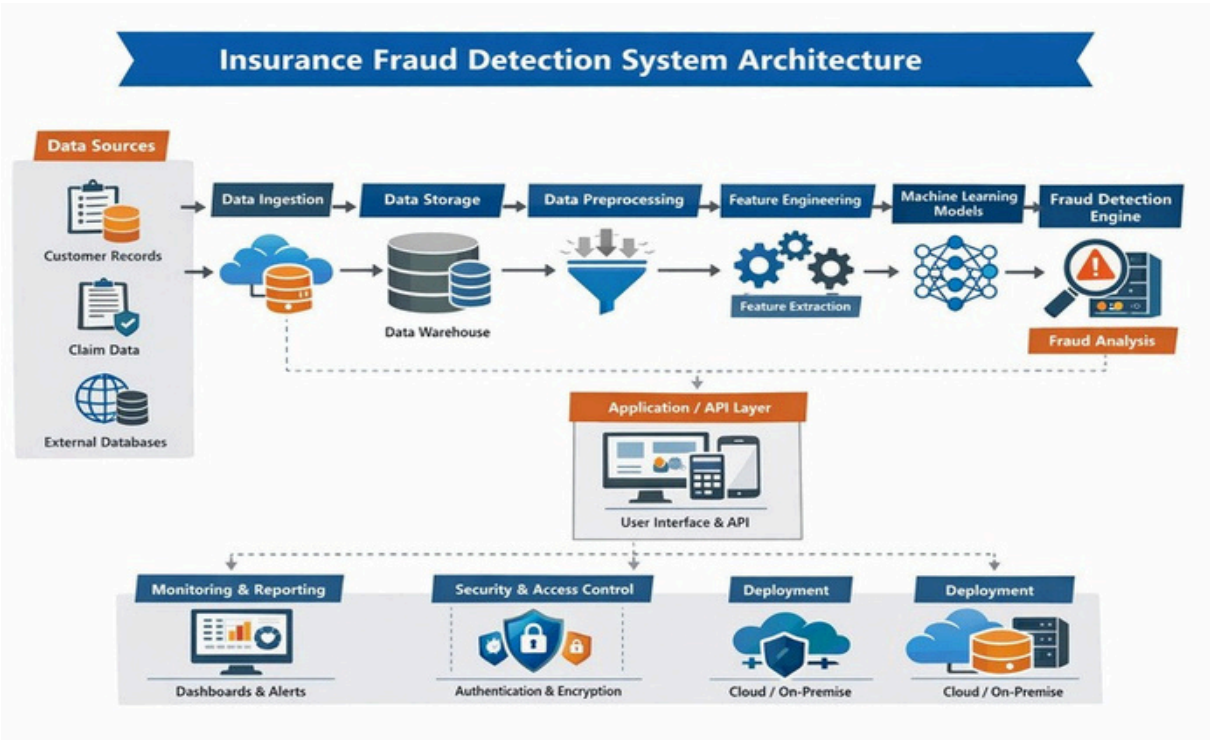


*Figure 1: Architecture and data flow of the voice patient diary sample application*